

Chapter 13.

$$13.1. (a) \quad \lambda_1 = 1 \quad \lambda_2 = -1$$
$$V_1 = C_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad V_2 = C_2 \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$(b) \quad \lambda_1 = 0 \quad \lambda_2 = 2$$
$$V_1 = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad V_2 = C_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

13.2. Proof. Since $\det A = \lambda_1 \lambda_2 \cdots \lambda_n$, we have.

$$A \text{ is singular} \Leftrightarrow \det A = 0 \Leftrightarrow \lambda_1 \lambda_2 \cdots \lambda_n = 0$$
$$\Leftrightarrow \lambda_i = 0 \text{ for some } 1 \leq i \leq n.$$

13.3. Proof. Let $A = \begin{bmatrix} a_{11} & & \\ a_{21} & a_{22} & \\ \vdots & \vdots & \ddots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is a lower triangular matrix.

$$\text{clearly, } A - \lambda I = \begin{bmatrix} a_{11} - \lambda & & \\ a_{21} & a_{22} - \lambda & \\ \vdots & \vdots & \ddots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{bmatrix}$$

$$\Rightarrow \det(A - \lambda I) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda)$$

$$\Rightarrow \lambda_1 = a_{11}, \lambda_2 = a_{22}, \dots, \lambda_n = a_{nn}$$

If A is an upper triangular matrix, the proof is similar.

$$13.4. \text{ For } \lambda = 3,$$
$$V = C_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda = -2$$
$$V = C_2 \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

13.5. (a). $\lambda_1 = 1$ $\lambda_2 = 3$ $\lambda_3 = -1$

(b) $\lambda_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $\lambda_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\lambda_3 = 1$.

(c) $\lambda_1 = 2$ $\lambda_2 = -1$ $\lambda_3 = 3$ $\lambda_4 = 1$

13.6. Proof: Suppose λ is an eigenvalue of M such that $|\lambda| \geq 1$.

Pick $V (\neq 0)$ which is an eigenvector corresponding to λ .

then $MV = \lambda V$.

$M^k V = \lambda^k V$

So $\lim_{k \rightarrow \infty} \|M^k V\| = \lim_{k \rightarrow \infty} \|\lambda^k V\| = \|V\| \lim_{k \rightarrow \infty} |\lambda|^k \geq \|V\|$.

which means $\lim_{k \rightarrow \infty} M^k V \neq 0$

13.7. For Jacobi method,

$S = \begin{bmatrix} 2 & \\ & 4 \end{bmatrix}$ $T = \begin{bmatrix} 0 & -3 \\ -1 & 0 \end{bmatrix} \Rightarrow S^{-1}T = \begin{bmatrix} 0 & -\frac{3}{2} \\ -\frac{1}{4} & 0 \end{bmatrix} \Rightarrow \lambda_1 = -\frac{\sqrt{6}}{4}$
 $\lambda_2 = -\frac{\sqrt{6}}{4}$

$\rho(S^{-1}T) = \max\{\frac{\sqrt{6}}{4}, \frac{\sqrt{6}}{4}\} = \frac{\sqrt{6}}{4} < 1$.

So the Jacobi method converges.

For Gauss-Seidel Method,

$S = \begin{bmatrix} 2 & \\ & 4 \end{bmatrix}$ $T = \begin{bmatrix} 0 & -3 \\ 0 & 0 \end{bmatrix} \Rightarrow S^{-1}T = \begin{bmatrix} 0 & -\frac{3}{2} \\ 0 & \frac{3}{8} \end{bmatrix} \Rightarrow \lambda_1 = 0$
 $\lambda_2 = \frac{3}{8}$

$\rho(S^{-1}T) = \max\{0, \frac{3}{8}\} = \frac{3}{8} < 1$

So the Gauss-Seidel method converges.

13.8. For Jacobi method,

$S^{-1}T = \begin{bmatrix} 0 & -2 & 2 \\ -1 & 0 & -1 \\ -2 & -2 & 0 \end{bmatrix} \Rightarrow \lambda_1 = 0$
 $\lambda_2 = 0$
 $\lambda_3 = 0 \Rightarrow \rho(S^{-1}T) < 1 \Rightarrow \text{Converges.}$

For Gauss-Seidel Method

$S^{-1}T = \begin{bmatrix} 0 & -2 & 2 \\ 0 & 2 & -3 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \lambda_1 = 0$
 $\lambda_2 = 2$
 $\lambda_3 = 2 \Rightarrow \rho(S^{-1}T) > 1 \Rightarrow \text{diverges}$

13.9. For Jacobi method,

$$S^{-1}T = \begin{bmatrix} 0 & -\frac{1}{2} & -1 \\ -1 & 0 & 0 \\ -1 & -1 & 0 \end{bmatrix} \Rightarrow \begin{aligned} \lambda_1 &= -1.4757 \\ \lambda_2 &= 0.7378 + 0.365i \\ \lambda_3 &= 0.7378 - 0.365i \end{aligned} \Rightarrow \rho(S^{-1}T) > 1 \Rightarrow \text{diverges.}$$

For Gauss-Seidel method.

$$S^{-1}T = \begin{bmatrix} 0 & -\frac{1}{2} & -1 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{aligned} \lambda_1 &= 0 \\ \lambda_2 &= \frac{1}{2} \\ \lambda_3 &= 0 \end{aligned} \Rightarrow \rho(S^{-1}T) < 1 \Rightarrow \text{Converges.}$$

13.10. Proof: $\det(A - \lambda I) = \det((A - \lambda I)^t)$
 $= \det(A^t - \lambda I^t) = \det(A^t - \lambda I)$

So A and A^t have identical eigenvalues.

13.11. Proof. Suppose V is an eigenvector of A corresponding to the eigenvalue λ , then

$$A^n V = A^{n-1}(AV) = \lambda A^{n-1}V = \dots = \lambda^n V$$

So λ^n is an eigenvalue of A^n and V is a corresponding eigenvector.

13.12. Trivial proof.

13.13. (a) $\sum_{i=1}^3 \lambda_i = 1+1+1 = 3 = \text{tr}(A)$

(b) $\sum_{i=1}^3 \lambda_i = 0+0+0 = 0 = \text{tr}(A)$

(c) $\sum_{i=1}^4 \lambda_i = 2-1+3+1 = 5 = \text{tr}(A)$

~~13.14. Proof: $\det(AB - \lambda I) = \det((AB - \lambda I)^t) = \det(B^t A^t - \lambda I)$~~

13.15. (a) $\rho(A) = 3$ (b) $\rho(A) = 1$ (c) $\rho(A) = 3$.